

Image

DINO-V2-Small
Vision Encoder

Caption
Text
Input

SmoLM-135M
Text Encoder

Multimodal Fusion

Language Model
Backbone

Final Logits

$\oplus$

LM Head

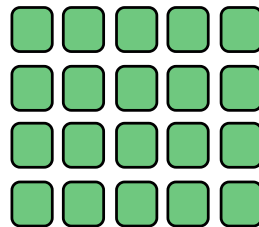
Feed-forward

Visual Self-Attention

Concatenate at Position

Episodic Memory
Module

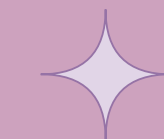
Found episodes
Add logit bias



$K \times C$
Memory Matrix

No found episodes
Write Episodes

Visual Absence
Refiner



Vision-Aware
Refiner

Final
Output
Sequence